

Lecture Notes: Risk and Return (Chapter 13)

Corporate Finance
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Preliminary Discussion & Midterm Syllabus

- The university's exam mode (online/physical) is still uncertain. The instructor emphasizes a lenient approach to exams given the challenges of the AI era.
- **Midterm Syllabus:**
 - Chapter 5: Time Value of Money (specifically the Amortization Table).
 - Chapter 6: Interest Rates.
 - Chapter 7: Bonds and their Valuation.
 - Chapter 13: Risk and Return (the topic of this lecture).

1 Introduction to Risk and Return

This chapter focuses on the risk and return of investments, primarily stocks. The main topics include:

- Expected Return
- Variance and Standard Deviation
- The impact of diversification
- The Systematic Risk Principle
- The Security Market Line (SML) and the Risk-Return Trade-off

2 Calculating Returns

Total return on a stock typically comes from two sources:

1. **Dividend Return:** Income received from the stock.
2. **Capital Gain:** The increase in the stock's price.

For a single-period investment horizon, we are concerned with the **expected return**.

2.1 States of the Economy and Probability

Returns are not guaranteed and vary with the state of the economy (e.g., Recession, Boom). Each state has an associated probability of occurrence.

Example: Stocks L and U

State of Economy	Probability	Stock L Return	Stock U Return
Recession	0.5	-20%	30%
Boom	0.5	70%	10%

2.2 Expected Return

The expected return is the probability-weighted average of the possible returns.

$$E(R) = \sum_{i=1}^n (p_i \times R_i)$$

where p_i is the probability of state i and R_i is the return in that state.

For Stock L:

$$E(R_L) = (0.5 \times -0.20) + (0.5 \times 0.70) = -0.10 + 0.35 = 0.25 = 25\%$$

For Stock U:

$$E(R_U) = (0.5 \times 0.30) + (0.5 \times 0.10) = 0.15 + 0.05 = 0.20 = 20\%$$

2.3 Risk Premium

The risk premium is the excess return an investor expects to receive for taking on additional risk. It is the difference between the expected return on a risky asset and the risk-free rate.

$$\text{Risk Premium} = E(R) - R_f$$

If the risk-free rate (R_f) is 8%, the risk premium for Stock L is $25\% - 8\% = 17\%$.

2.4 Variance and Standard Deviation

Variance (σ^2) and standard deviation (σ) measure the volatility or risk of an investment. They quantify the deviation of possible returns from the expected return.

$$\sigma^2 = \sum_{i=1}^n p_i \times (R_i - E(R))^2$$

$$\sigma = \sqrt{\sigma^2}$$

For Stock L:

$$\begin{aligned}\sigma_L^2 &= 0.5 \times (-0.20 - 0.25)^2 + 0.5 \times (0.70 - 0.25)^2 \\ &= 0.5 \times (-0.45)^2 + 0.5 \times (0.45)^2 \\ &= 0.5 \times 0.2025 + 0.5 \times 0.2025 = 0.2025 \\ \sigma_L &= \sqrt{0.2025} = 0.45 = 45\%\end{aligned}$$

For Stock U:

$$\begin{aligned}\sigma_U^2 &= 0.5 \times (0.30 - 0.20)^2 + 0.5 \times (0.10 - 0.20)^2 \\ &= 0.5 \times (0.10)^2 + 0.5 \times (-0.10)^2 \\ &= 0.5 \times 0.01 + 0.5 \times 0.01 = 0.01 \\ \sigma_U &= \sqrt{0.01} = 0.10 = 10\%\end{aligned}$$

Interpretation: Stock L has a higher expected return (25%) but also much higher risk (45% standard deviation) compared to Stock U (20% return, 10% risk).

2.5 Coefficient of Variation (CV)

The CV measures risk per unit of expected return. It is useful for comparing investments with different expected returns.

$$CV = \frac{\sigma}{E(R)}$$
$$CV_L = \frac{45\%}{25\%} = 1.8 \quad CV_U = \frac{10\%}{20\%} = 0.5$$

A lower CV indicates better risk-adjusted returns. Stock U is preferable by this measure.

Practice Question

State of Economy	Probability	Stock Return
Weak	0.1	-50%
Below Average	0.2	-5%
Average	0.4	16%
Above Average	0.2	25%
Strong	0.1	60%

Task: Calculate the Expected Return, Variance, Standard Deviation, and Coefficient of Variation.

Solution:

$$\begin{aligned}
 E(R) &= (0.1 \times -0.5) + (0.2 \times -0.05) + (0.4 \times 0.16) \\
 &\quad + (0.2 \times 0.25) + (0.1 \times 0.60) \\
 &= -0.05 - 0.01 + 0.064 + 0.05 + 0.06 = 0.114 = 11.4\%
 \end{aligned}$$

$$\begin{aligned}
 \sigma^2 &= 0.1(-0.5 - 0.114)^2 + 0.2(-0.05 - 0.114)^2 \\
 &\quad + 0.4(0.16 - 0.114)^2 + 0.2(0.25 - 0.114)^2 \\
 &\quad + 0.1(0.60 - 0.114)^2 \\
 &= 0.1(-0.614)^2 + 0.2(-0.164)^2 + 0.4(0.046)^2 \\
 &\quad + 0.2(0.136)^2 + 0.1(0.486)^2 \\
 &= 0.1(0.376996) + 0.2(0.026896) + 0.4(0.002116) \\
 &\quad + 0.2(0.018496) + 0.1(0.236196) \\
 &= 0.0376996 + 0.0053792 + 0.0008464 \\
 &\quad + 0.0036992 + 0.0236196 = 0.071244
 \end{aligned}$$

$$\sigma = \sqrt{0.071244} = 0.2669 = 26.69\%$$

$$CV = \frac{26.69\%}{11.4\%} = 2.34$$

3 Portfolio Expected Return

A **portfolio** is a group of assets held by an investor. **Portfolio weights** (w_i) represent the percentage of the total portfolio value invested in a particular asset.

Example: Portfolio of L and U (50-50 weights)

$$E(R_p) = w_L E(R_L) + w_U E(R_U) = 0.5 \times 0.25 + 0.5 \times 0.20 = 0.125 + 0.10 = 0.225 = 22.5\%$$

More generally, for a portfolio with n assets:

$$E(R_p) = \sum_{i=1}^n w_i E(R_i)$$

where $\sum_{i=1}^n w_i = 1$.

Example: Three-Stock Portfolio

State	Probability	Stock A	Stock B	Stock C
Boom	0.4	10%	15%	20%
Bust	0.6	8%	4%	0%

Individual expected returns:

$$E(R_A) = 0.4(10) + 0.6(8) = 8.8\%$$

$$E(R_B) = 0.4(15) + 0.6(4) = 8.4\%$$

$$E(R_C) = 0.4(20) + 0.6(0) = 8\%$$

If equally weighted ($w_A = w_B = w_C = 1/3$):

$$E(R_p) = \frac{1}{3}(8.8) + \frac{1}{3}(8.4) + \frac{1}{3}(8) = 8.4\%$$

4 Next Class

We will cover the calculation of portfolio variance and standard deviation, which is not simply a weighted average of individual variances.